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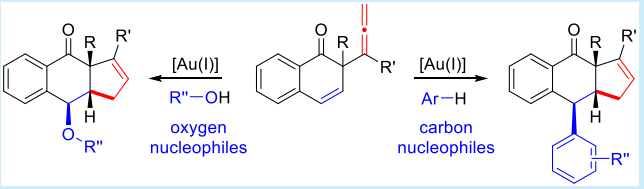
Controlling the Gold(I)-Catalyzed 1,5-Allenene Reaction: Construction of Fused Rings with Excellent Diastereoselectivity

Nina Semleit and Gebhard Haberhauer*

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ABSTRACT: In the present study, the gold(I)-catalyzed reaction of 1,5-allenenes was controlled in such a way that instead of a [2 + 3] cycloaddition, a 5-*exo*-cyclization with the formation of a carbocation occurred. The latter could be trapped with both oxygen and carbon nucleophiles. In the investigated system, fused tricyclic frameworks with three contiguous stereocenters with excellent chemo- and diastereoselectivity in up to 95% yield were obtained.



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The usage of gold complexes for the electrophilic activation of carbon−carbon multiple bonds for attack by various nucleophiles under mild conditions has been intensively investigated over the past decade.[1−12] In addition to alkynes,[8] allenes represent one of the most frequently used substance classes in the field of homogeneous gold catalysis.[5,12−18] For example, an intramolecular reaction of an allene unit with heteronucleophiles such as oxygen or nitrogen derivatives leads to a series of heterocyclic compounds.[16,17] In the case of allenenes, the gold-activated allene unit attacks the double bond of the alkene moiety, and an alkyl cation is formed.[14,19,20] The fate of this species depends on the relative position of the allene and alkene units within the molecule, the substituents at the multiple bonds, and the reaction conditions; that is, the reaction could be terminated by either deprotonation or nucleophilic attack. In general, the nucleophile is a carbon atom of the allenic unit leading to a [2 + 2] or [2 + 3] carbocycloaddition.[14] Whereas interception of the intermediate carbocation by heteronucleophiles was possible in the case of 1,7-allenenes,[21] in nonactivated 1,5-allenenes, the cationic intermediate **Int1**, which is formed through 6-*endo*-cyclization, cyclizes to the gold carbene **Int2** (Scheme 1a).[20,22,23] After removal of the gold catalyst, bicycle **2** is obtained. Depending on the substituent pattern, **2** can undergo further gold-catalyzed cyclization reactions, leading to polycycles.[20,23] In this work, we intended to favor the 5-*exo*-cyclization by introducing an aromatic moiety stabilizing the benzylic cation **Int3**. The latter could be transferred to the cyclopentene derivatives **4** by an intermolecular nucleophilic attack of both oxygen and carbon nucleophiles (Scheme 1b).

We have chosen the bicyclic allenenes **5** as model systems for two reasons. On the one hand, these systems can be easily prepared from alkynylnaphthol ethers by gold catalysis.[24] On the other hand, the desired products **6** represent 6−6−5-fused rings with multiple stereogenic centers, which are of great interest because this structural motif is frequently found in natural products and pharmaceutical compounds with important biological properties. Examples are picropodophyllone, (−)-epipodophyllotoxin, (−)-podophyllotoxin, and steronstrein A[25] (Scheme 1c). The latter is a terpenoid isolated from *Stereum ostrea* and shows antimalarial and cytotoxic activity.[25] Picropodophyllone, (−)-epipodophyllotoxin, and (−)-podophyllotoxin are naturally occurring aryltetralin cyclolignans belonging to the family of podophyllotoxins.[26] They exhibit various biological activities ranging from insecticidal, antifungal, antiviral, anti-inflammatory, and neurotoxic activities to immunosuppressive and further properties.[26] The class of podophyllotoxins is characterized by the presence of a carbonyl or hydroxyl group and an electron-rich aromatic compound in the 1,4-position at the central six-membered ring. Because (−)-epipodophyllotoxin is the aglycone of the antitumor drugs etoposide and teniposide,[27−29] there have been considerable efforts in developing methods to construct such fused tricyclic frameworks.[30−33] The design of alternative synthetic routes to diverse tricyclic scaffolds with different functionalities and stereocenters is therefore highly desired.

Previous work demonstrated, that 1,5-allenedienes and 1,5-allenenes can be prepared in only one step via the Au-catalyzed rearrangement of propargyl aryl ethers.[24,34] Because we have some experience with the gold-catalyzed reactions of haloacetylenes,[35−38] we have chosen chloroacetylene **7a** as the starting material, which should react toallenene **5a** via a gold(I)-catalyzed rearrangement (Table 1). We then investigated the gold(I)-catalyzed reaction of **7a** with[JohnPhosAu-

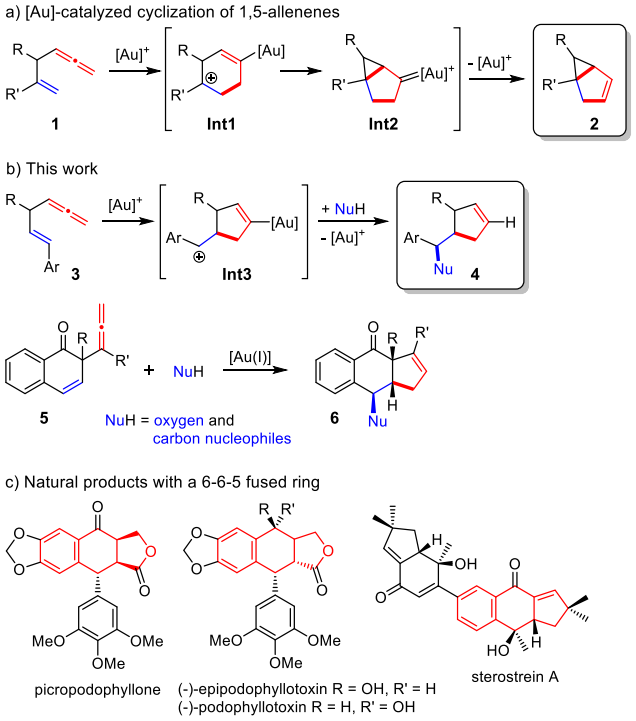
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Scheme 1[a]



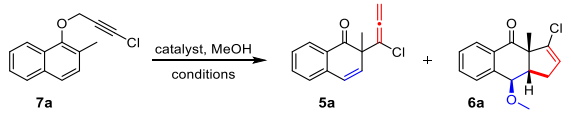
[a](a) Au(I)-catalyzed cyclization of 1,5-allenenes leads to bicycle **2**, which can react under the given conditions to form polycycles. (b) Presented Au(I)-catalyzed cyclization of 1,5-allenenes via benzylic intermediate **Int3**, which can be intercepted by oxygen (alcohol and phenol) and carbonucleophiles (alkenes and arenes). (c) Representative natural products exhibiting a 6−6−5-fused ring.

(NCMe)]SbF_6[39] and methanol. Fortunately, we were able to detect a single product with a yield of 87% after only 2 h of reaction time (entry 1). Nuclear magnetic resonance (NMR) spectroscopic studies revealed that, in fact, the 6−6−5 cycle **6a** was obtained. The intermediary formed allenene **5a** could not be observed after this reaction time.

Subsequently, we tested a wide variety of phosphine ligands to improve the yield, which was not successful (entries 2−7). The usage of a carbene ligand did not lead to a better yield despite varying the counterions (entries 8−10). The catalyst dichloro(2-picolinato)gold(III)[40] exclusively lead to the formation of **5a** in 88% yield (entry 11). Because the ligand JohnPhos achieved the highest yield, we subsequently varied the counterion[41] of this catalyst. Again, no improvement in the yield was achieved (entries 12−14). The usage of chloroform as a solvent proved to be the most suitable compared with DCM and DCE (entries 15 and 16). From these results, we concluded that an optimum yield is obtained with a ratio of **7a** (0.1 M) to methanol of 1:2, the catalyst [JohnPhosAu(NCMe)]SbF_6 (5 mol %) in chloroform at room temperature, and a reaction time of 2 h. Using these conditions, the reaction can also be performed on a millimole scale.

To demonstrate that the reaction proceeds via allenene **5a** as an intermediate, we isolated the latter starting from acetylene **7a** with dichloro(2-picolinato)gold(III) in 98% yield (Scheme 2). Analogously, we were able to obtain allenenes **5b−d** in good to excellent yields. If the allenenes **5a−d** were now reacted in a second step with methanol and [JohnPhosAu(NCMe)]SbF_6, then cycles **6a−d** were obtained (Scheme 3). Accordingly, allenenes **5** were the starting materials of the

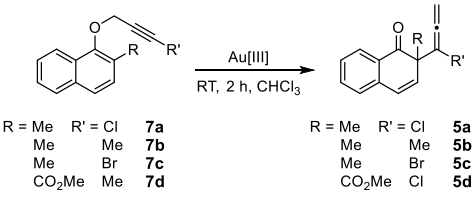
Table 1. Optimization of the Reaction Conditions for the Gold-Catalyzed (5 mol % Catalyst and 7 mol % Counterion) Formation of 5a and 6a Starting from Alkyne 7a and Methanol (2.0 equiv) at Room Temperature in Chloroform for 2 h



		yield (%)	
entry	catalyst	**5a**	**6a**
1	[JohnPhosAu(NCMe)]SbF_6		87
2	XPhosAu(NCMe)SbF_6		78
3	BrettPhosAuNTf_2		80
4	^{t}BuXPhosAuNTf_2		79
5	CyJohnPhosAuCl/AgSbF_6		76
6	Ph_3PAuCl/AgSbF_6		65
7	Me_3PAuCl/AgSbF_6	57	
8	IPrAuCl/AgNTf_2		55
9	IPrAuCl/NaBArF_{24}	36	31
10	IPrAuCl/AgSbF_6		61
11	Dichloro(2-picolinato)gold(III)	88	
12	JohnPhosAuCl/AgSbF_6		75
13	JohnPhosAuCl/AgNTf_2		64
14	JohnPhosAuCl/NaBArF_{24}	46	36
15[a]	[JohnPhosAu(NCMe)]SbF_6		82
16[b]	[JohnPhosAu(NCMe)]SbF_6		53
17[c]	[JohnPhosAu(NCMe)]SbF_6		56
18[d]	[JohnPhosAu(NCMe)]SbF_6		84
19[e]	[JohnPhosAu(NCMe)]SbF_6		83
20[f]	[JohnPhosAu(NCMe)]SbF_6	60	22

[a]DCM. [b]DCE. [c]1 equiv of MeOH. [d]3 equiv of MeOH. [e]2.5 mol % catalyst. [f]1 mol % catalyst.

Scheme 2. Synthesis of the 1,5-Allenenes 5 via Au(III)-Catalyzed Rearrangement Starting from the Readily Available Naphthol Derivatives 7

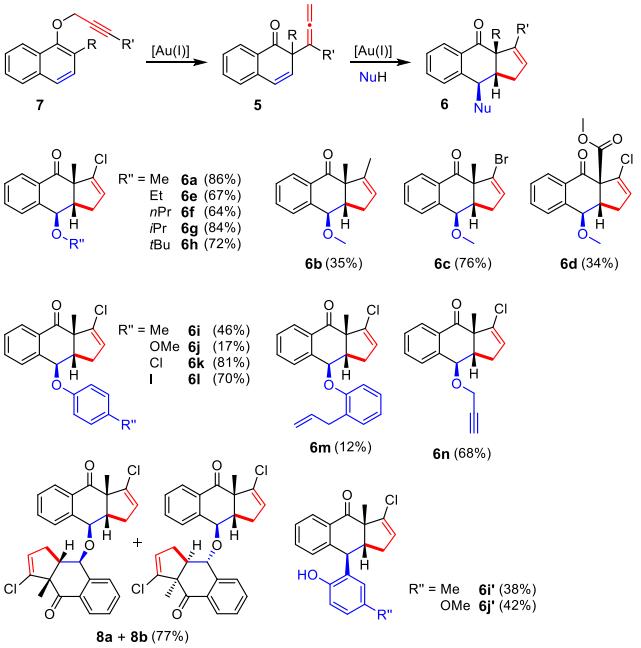


subsequent cyclization. Because the yields of cycles **6a−d** are independent of whether the reaction starts from the acetylenes **7a−d** or the intermediate allenenes **5a−d**, acetylenes **7a−d** are often directly converted to avoid the isolation step of the corresponding allenenes.

Next, we wanted to evaluate the scope of the reaction by varying the substituents and the oxygen nucleophile. Different substituents on the alkyne unit of **7** were tolerated (**6b−c**). The introduction of an electron-withdrawing substituent at the two-position of the naphthol scaffold also led to the desired cycle **6d**. Using the terminal alkyne (R′ = H), a rearrangement of the propargylic unit occurred (Scheme S1). Starting from chloroacetylene **7a** and different aliphatic alcohols, we were able to isolate the corresponding ethers (**6a** and **6e−6h**) in good to very good yields. The introduction of phenols as oxygen nucleophiles was successful, too (**6i−6m**). Using

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Scheme 3. Evaluation of the Substrate Scope of the Gold(I)-Catalyzed 1,5-Allenene Cyclization Using Oxygen Nucleophiles[a]



[a]Allenenes **5** were generated in situ from the corresponding alkynes **7**. The products were isolated as racemic mixtures. Only one of the enantiomers is shown in each case. Reaction conditions: RT, 3 h, $CHCl_3$.

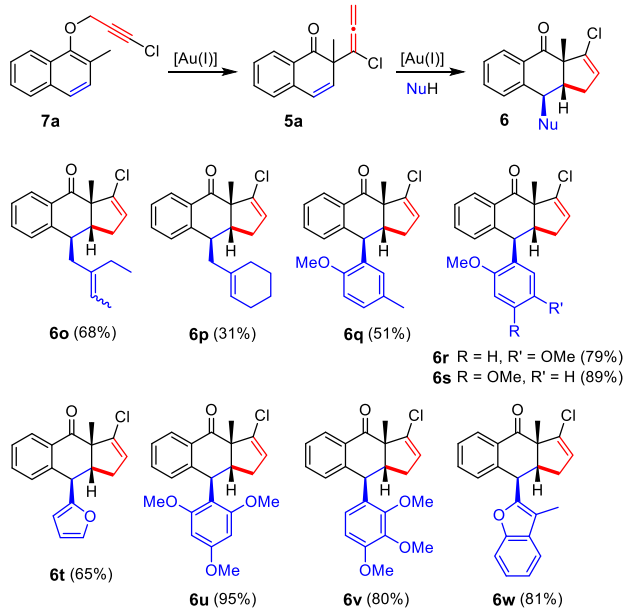
propargyl alcohol as a nucleophile resulted in the formation of cycle **6n** with a yield of 68%. Interestingly, the cyclization and subsequent reaction with the oxygen nucleophile proceeded with excellent stereoselectivity. Only one of the possible diastereomers could be isolated as an enantiomeric pair in all cases. The stereochemistry of the product could be determined by means of nuclear Overhauser effect spectroscopy (NOESY) of the cycles **6e** and **6g** (Figures S7 and S8). Upon introducing a stereogenic center to the propargylic unit (**7e**), we were able to isolate two diastereomers (**6x** and **6x′**; see Scheme S2) with a ratio of 60:40. Thus the stereogenic center at the propargylic unit barely influenced the diastereoselectivity of the cyclization reaction.

We were interested in the reaction with [JohnPhosAu(NCMe)]SbF_6 as a catalyst without the addition of a nucleophile. Here the question arises of whether the benzylic cation reacts intramolecularly to form a bicycle, as in the case of the Au-catalyzed reactions of 1,5-allenenes described so far. (See Scheme 1a.) As a result, a mixture of two diastereomers was isolated in good yields and identified as ethers **8a** and **8b** by mass spectrometry (Scheme 3). They are probably formed by the reaction of the benzylic cation with traces of water present in the solvent. The thus-formed alcohol acts as a nucleophile in the reaction with another benzylic cation.

Overall, yields of the cyclization were good to very good. Particularly striking, however, are the low, <20% yields of cycles **6j** and **6m**. Whereas in the case of **6m** the additional carbon nucleophile (double bond of the alkene) presumably leads to a polymerization without isolable byproducts, a more detailed analysis of the reaction of **7a** with 4-methoxyphenol shows that a second product (**6j′**) is formed. Thus both the oxygen of the phenol and a carbon atom of the aromatic system can act as a nucleophile. Analogous observations were made for the usage of cresol as a nucleophile (**6i** and **6i′**).

These results encouraged us to employ carbon nucleophiles to trap the intermediary formed benzylic cation. Indeed, we were successful by using alkenes and electron-rich aromatic systems (Scheme 4). Both react under addition, subsequent

Scheme 4. Evaluation of the Substrate Scope of the Gold(I)-Catalyzed 1,5-Allenene Cyclization Using Carbon Nucleophiles[a]



[a]Allenene **5a** was generated in situ from the corresponding alkyne **7a**. The products were isolated as racemic mixtures. Only one of the enantiomers is shown in each case. Reaction conditions: RT, 3 h, $CHCl_3$.

deprotonation, and protodesauration. In the case of the alkenes, this formally results in migration of the double bond, leading to regioisomers (**6o**). Note that in all cases, the addition of the nucleophile is highly stereoselective. For aromatic compounds, Friedel–Crafts alkylation is carried out according to the rules of second substitution on the aromatic ring with the following tendency: The more electron-rich the aromatic system, the higher the yields. For example, yields of 95% can be obtained using 1,3,5-trimethoxybenzene (**6u**). As in the case of the oxygen nucleophiles, only one diastereomer can be isolated as an enantiomeric pair (Figure S9).

To investigate the mechanism of the cyclization, we performed quantum-chemical calculations by means of B3LYP-D3BJ(SMD).[42−45] The results are summarized in the catalytic cycle, which is shown in Figure 1a. First, the gold catalyst coordinates to the allene unit of **5a**, resulting in the gold complex **Int4**. The latter is considered as a reference with an energy of 0.0 kcal/mol. From there, two different routes are possible. On the one hand, a [2 + 3] cycloaddition can take place with the formation of the polycycle **Int6**. On the other hand, cyclization is possible, yielding the formation of a five-membered ring (**Int5**). A closer look at the calculated structure of **Int5** reveals a T-shaped edge-to-face $\pi-\pi$ interaction between the *o*-phenyl group of JohnPhos and the substrate (Figure 1b). The distance between the centers of the aromatic units amounts to 4.95 Å.[46] A comparison of the free energies

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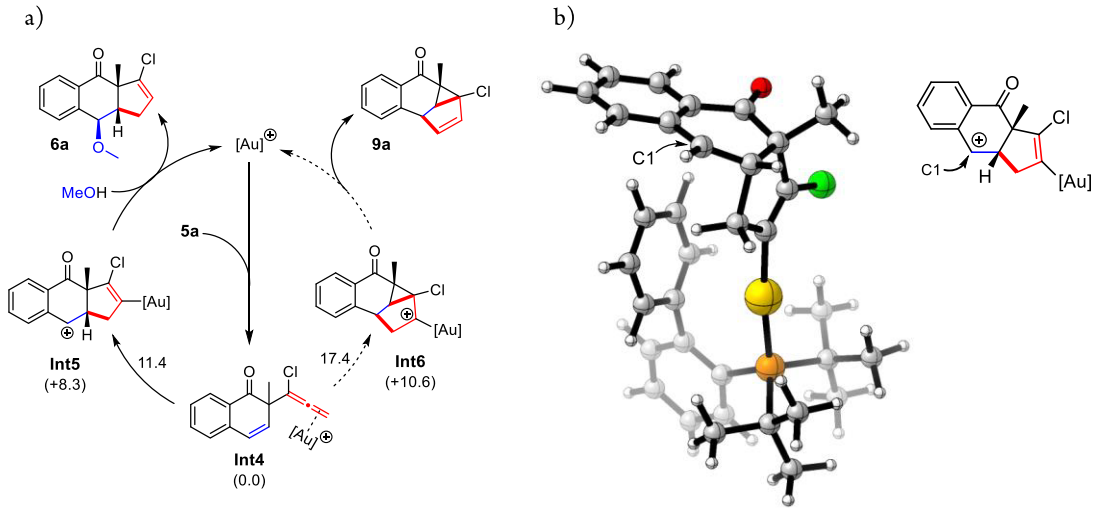


Figure 1. (a) B3LYP-D3BJ(SMD)-calculated catalytic cycle for the gold(I)-catalyzed 1,5-allenene cyclization of **5a**. The free energies (ΔG) of the intermediates **Int4**–**Int6** (numbers in brackets) and transition states (numbers above the arrows) are given in kcal/mol and are relative to **Int4**. (b) Molecular structure of **Int5** calculated by means of B3LYP-D3BJ. $[Au]^+$ = JohnPhosAu$^+$.

(ΔG) of the transition states shows that the 5-*exo*-cyclization is significantly more favored than the [2 + 3] cycloaddition (+11.4 vs +17.4 kcal/mol). Therefore, only the 5-*exo*-cyclization bearing an exocyclic cation occurs. The latter is stabilized by the adjacent aromatic system and can react with a nucleophile to form the product **6a**. The stereoselectivity of the reaction can be explained by looking more closely at **Int5**, where the catalyst shields one side of the cation (C1 atom in Figure 1b). Thus the attack of the nucleophile is possible only from the opposite side, which explains the excellent stereoselectivity. Calculations using the small nucleophile MeOH show an energy difference of the transition states (nucleophilic attack from the top and the bottom) of >8 kcal/mol. This corresponds to an exclusive reaction from the upper side, which was also observed in the experimental procedures.

In conclusion, we have successfully demonstrated that the gold(I)-catalyzed reaction of nonactivated 1,5-allenenes can be controlled. Normally, this reaction leads, via [2 + 3] cycloaddition, to bicyclic and, in further steps, to polycyclic compounds. We showed that the introduction of an aromatic substituent leads to a 5-*exo*-cyclization with the formation of a carbocation. The latter can react with both oxygen and carbon nucleophiles. Starting from a bicyclic system, which can be prepared in situ from a naphthol derivative, 6–6–5-fused rings exhibiting three stereocenters with excellent chemo- and diastereoselectivity were obtained. Because of the very good yields, this reaction provides a new facile and straightforward approach for the diversity-oriented synthesis of such 6–6–5-fused rings.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03886.

Molecular structures, synthesis of the new compounds, NMR spectra, computational details, Cartesian coordinates and absolute energies for all calculated compounds, HRMS data, and ^{1}H NMR and ^{13}C NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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